ALERT meeting

Efficiency of the time reconstruction of saturated signals

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IJCLab

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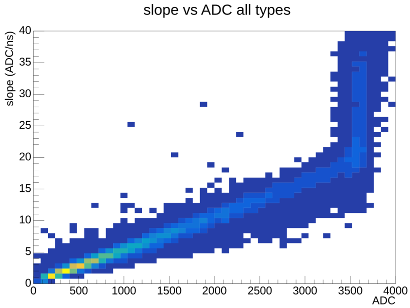
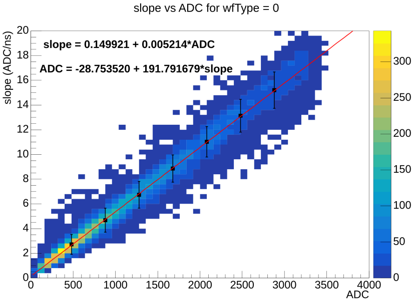
- The amplitude of saturated signals can be estimated using

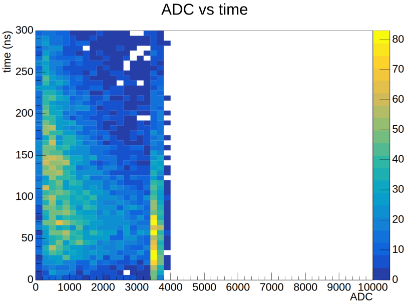
$$ADC_{corr} = f(slope)$$

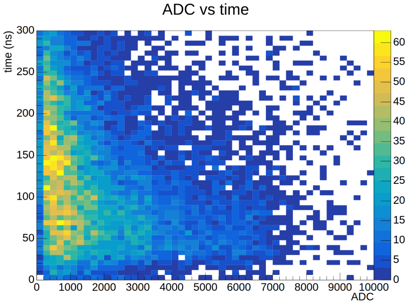
where f is extracted from non-saturated signals





- Once we have a reasonable estimate of the amplitude, we can re-compute the time at half amplitude
- We no longer have the accumulation of small time at 3500 ADC

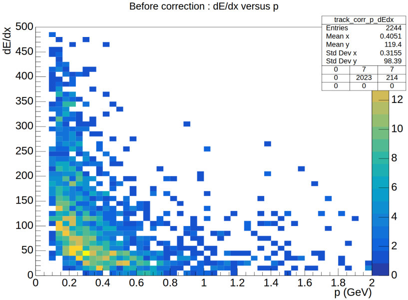


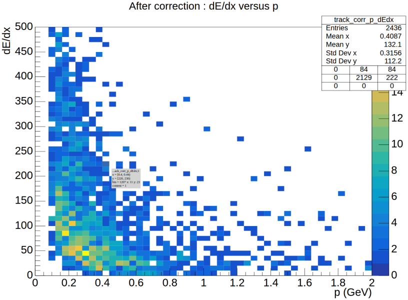


Results 2/2

- We have earned some tracks.
- Stats for run 21476, 4He, 10.6 GeV
- Cuts: 1 electron, 1 track with nhits $>=7$ and chi2 <8

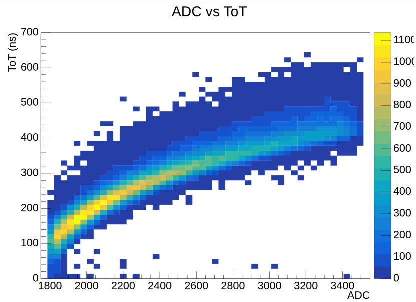
	coatjava dev	after adc correction
nb. tracks	2244	2436

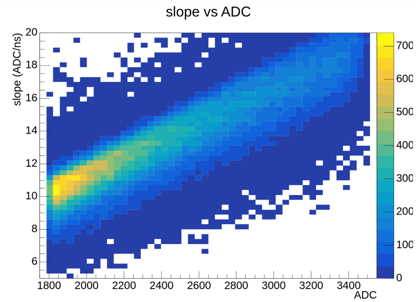




Other approach

- I selected good signals with $1800 < ADC < 3500$
- I compute the ToT at a fixed threshold : 2000
- For the same samples, I plot ADC versus ToT and versus Slope
- The ToT seems more precise than the Slope, but it is not linear!
- How should we extrapolate the ADC versus ToT relation ?
- I plan to study the ADC resolution for each method.





- Bias in the time reconstruction of saturated signals
- For such signals, the arrival time and the timeOverThreshold can be wrong